vapour, the molecules dart about erratically, in all directions, like a swarm of midges, at considerable distances from each other, and in a state of complete disorder. Their movement is closely dependent upon the temperature of the vapour; heat is always associated with a disorderly motion of the molecules. In the case of the larger particles, a powerful microscope will enable us to discern the so-called *Brownian movement*, which becomes always more and more marked as the temperature of the substance increases. In liquids, the molecules are likewise in a state of disorder, but they are closely packed and move in between each other, in a manner comparable to the movement of ants in an ant-hill. In crystalline, solid matter, the atoms, or molecules, are likewise tightly packed, but here they constitute a completely orderly pattern. Figure 2 shows the model of a rock salt crystal. Rock salt is a chemical compound of the elements chlorine and sodium. In this model, the black dots stand for chlorine atoms, the white dots for sodium atoms. They alternate in the crystal in a perfectly regular pattern. Actually, these atoms are in a state of more or less violent movement, according to the temperature of the crystal; they vibrate about their positions of equilibrium. Since the atoms are actually packed tightly, without any empty space between them, the above model does not exactly correspond to reality.

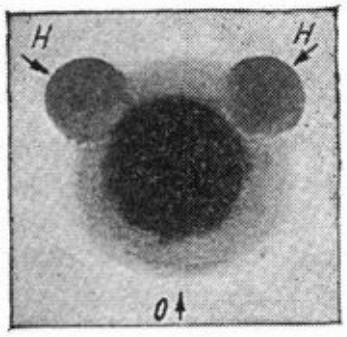


Figure 1.—Model of the molecule of water.

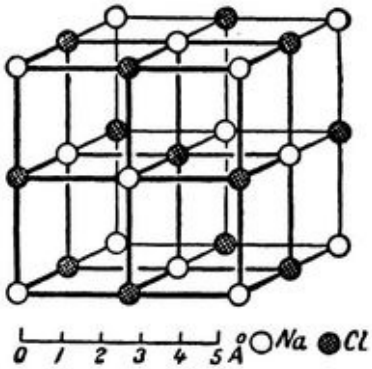


Figure 2.—Model of a common salt crystal.

The question now is: What do such schematic models actually mean, and